Starved in the womb: The Biafran blockade and men's schooling

Glory David Adebayo* Jean Msabila†

October 2026

Abstract

Wartime famines are rarely separable from the wars that produce them. We exploit the timing of the 1968–70 Biafran blockade, which began eleven months into the Nigerian Civil War, and exact month of birth in five Nigeria Demographic and Health Surveys. Men gestated during the famine complete 1.3 fewer years of schooling and are 12 percentage points less likely to reach secondary school; women are unaffected. The estimate exceeds all 74 placebo estimates, holds in a triple difference against women, and survives controls for Universal Primary Education. Starvation in war leaves a lasting deficit in human capital.

JEL classification: I15, I21, J16, N37, O15
Keywords: Famine; Fetal origins; Schooling; Civil war; Nigeria

*Corresponding author. Department of Economics, Florida International University, Miami, FL 33199, USA. E-mail: gadeb001@fiu.edu.

†Department of Economics, West Virginia University, Morgantown, WV 26506, USA. E-mail: jam00124@mix.wvu.edu.

1 Introduction

Starvation remains an instrument of war. From Yemen and Tigray to Sudan and Gaza, sieges and restrictions on food have placed entire populations at risk of famine (de Waal, 2017). The humanitarian toll is immediate, but the economic costs may be far more durable: adverse conditions in utero can permanently lower health and human capital (Almond and Currie, 2011; Currie and Vogl, 2013). Yet the long-run consequences of wartime famine are difficult to isolate, because famines rarely occur apart from the violence and dislocation of the wars that produce them.

This paper exploits a setting in which the famine can be separated from the war. The Nigerian Civil War began in July 1967, but Biafra did not starve until federal forces captured Port Harcourt in May 1968 and sealed the enclave; the blockade famine then lasted until Biafra's surrender in January 1970 (Heerten and Moses, 2014). For eleven months, Biafra was at war but not yet starving. Using exact month of birth for adults in five pooled rounds of the Nigeria Demographic and Health Surveys (NDHS), we compare cohorts whose gestation overlapped the famine with neighboring cohorts, across ethnic groups that were and were not exposed to the war, and separately estimate the effect of gestation during the war months that preceded it.

We find that the famine left a large and lasting mark on the schooling of boys who were in utero during it. Men gestated during the famine complete 1.3 fewer years of schooling, are 6.6 percentage points less likely to have ever enrolled, and are 12 percentage points less likely to reach secondary school. Women gestated during the famine show no loss. This male-specific pattern accords with the vulnerability of male fetuses to maternal undernutrition (Kraemer, 2000) and is difficult to reconcile with explanations based on household resources or school supply, which would affect both sexes.

The result withstands demanding falsification tests. It lies outside the distribution of 74 placebo famines, is negative in every survey round, and is nearly identical in a triple difference against women of the same groups and cohorts. It is also robust to the main

contemporaneous shock, the 1976 introduction of Universal Primary Education (UPE).

This paper makes three contributions. First, to our knowledge, it is the first study to use timing within a war to isolate the long-run effects of wartime famine. Existing studies treat the conflict's 31 months as a single shock (Akresh et al., 2012, 2023; Ajefu and Casale, 2021), and public-health studies of Biafran cohorts do not exploit timing within the war (Hult et al., 2010; Ogah et al., 2023). Second, it extends evidence on the long-run effects of famine and conflict (Almond et al., 2010; Meng and Qian, 2009; Neelsen and Stratmann, 2011; Scholte et al., 2015; Akbulut-Yuksel, 2014) to a large African wartime famine and documents a male-specific effect on educational attainment. Third, it shows that the ethnicity-by-cohort design underlying prior estimates fails a placebo war test once survey rounds are pooled, underscoring the value of within-war variation.[1]

2 The Biafran war and blockade

Biafra declared independence on 30 May 1967, uniting the Igbo heartland with the Ibibio, Ekoi, and Ijaw territories of the former Eastern Region. Federal forces took Enugu and Calabar in October 1967, Onitsha in March 1968, and Port Harcourt in May 1968, leaving Biafra a landlocked enclave. From June 1968, food and relief were severely restricted; by August, relief agencies described the crisis as the gravest humanitarian emergency since the Second World War, and night airlifts met only a fraction of needs.[2]

Famine-gestated children reached school age in 1975–76, just as Nigeria introduced UPE, which abolished primary school fees nationwide (Csapo, 1983; Osili and Long, 2008). Because Northern enrollment started from a very low base, UPE generated its largest gains in the North, among cohorts born in precisely the years the famine struck the East; Section 5 addresses this coincidence directly.

[1] Online Appendix Table A11 re-estimates that design for several of the literature's outcomes.
[2] Online Appendix C provides a chronology of the war and the famine.

3 Data

We use the individual recode files of the 2003, 2008, 2013, 2018, and 2024 NDHS, which record each respondent's month and year of birth. The sample comprises respondents born between January 1965 and December 1973 with exact, non-imputed birth dates.[3] Because ethnicity is coded differently across rounds, we define the treated group as the four broad groups that can be identified consistently and whose homelands lay inside Biafra in 1967: the Igbo, Ijaw, Ibibio, and Ekoi, who account for 69, 17, 10, and 4 percent of treated men. Following Akresh et al. (2012), we assign exposure by ethnicity rather than current residence, which was itself shaped by the war and subsequent migration. Since parts of the minority homelands fell before the famine, the estimates are intent-to-treat effects; restricting the treated group to the Igbo leaves them essentially unchanged.

The outcome, completed years of schooling, is fixed by the time respondents are surveyed at ages 29–59. The sample contains 8,327 men and 15,185 women, including 1,785 men and 3,747 women from the war-affected groups; 436 of these men spent at least one month of gestation in the famine. Online Appendix Table A1 reports descriptive statistics.

4 Empirical model

For a respondent born in month b, let F_b and P_b denote the number of gestation months (from $b-9$ to $b-1$) that overlapped the famine (June 1968–January 1970) and the pre-famine war (July 1967–May 1968), each ranging from zero to nine. We estimate

$$\begin{aligned} Y_{iebsr} = \beta_F \left(War_e \times F_b \right) + \beta_P \left(War_e \times P_b \right) + \mathbf{X}'_{eb}\gamma + \delta \left(War_e \times t_b \right) \\ + \mu_b + \eta_e + \lambda_{sr} + \kappa_{War_e,a} + \omega_{War_e,m} + \varepsilon_{iebsr}, \end{aligned} \tag{1}$$

[3] Among war-ethnicity men, the share with imputed birth months falls from 18 percent in 2003 to zero in 2024. Including imputed dates yields similar estimates (Table 2). Variable definitions are in Online Appendix B.

where War_e indicates the war-affected groups, $\mathbf{X}_{eb}$ contains the analogous interactions for months of infancy (ages 0–23) in each period, and $\delta(War_e \times t_b)$ is a war-group linear cohort trend. Birth year-month fixed effects μ_b absorb everything common to Nigerians born in the same month; η_e and λ_{sr} are ethnicity and state-by-round fixed effects; and $\kappa_{War_e,a}$ and $\omega_{War_e,m}$ give the war groups their own five-year age profile, identified because pooled rounds observe each cohort at different ages, and their own calendar-month-of-birth effects. We report $9\beta_F$, the effect of full gestation during the famine, weight by DHS sampling weights normalized within round, and cluster standard errors by ethnicity × birth year. The specification and primary outcomes were fixed before estimation.[4]

The identifying assumption is that, absent the famine, the schooling gap between war and non-war groups born in the same month would have evolved smoothly across cohorts. Because every war-ethnicity child born before 1967 lived through the war in childhood, there is no untreated pre-period, and pre-trend tests have limited power in any case (Roth, 2022). We therefore assess the assumption with event studies, a triple difference by sex, placebo famines, and UPE controls. Figure 1 plots the war–non-war gap by year of birth relative to cohorts born in 1971–72, who were never exposed. For men, the gap is stable before and during the early war and falls by about one year for cohorts born in 1969 and 1970, whose gestation overlapped the famine. For women, it rises gradually, with no break. Figure 2 makes the comparison explicit: once the path common to both sexes is removed, the men-minus-women coefficients are centered on zero for every cohort born before the famine and drop to about −1.0 in 1969 and 1970.[5]

5 Empirical results

Main result. Table 1 reports the main estimates. Full gestation during the famine reduces men's completed schooling by 1.3 years ($p = 0.004$), or 15 percent of the sample mean. The

[4]They were fixed in writing but not publicly pre-registered. The UPE controls and comparison-group splits in Section 5 were added afterward to address specific threats.

[5]Online Appendix Figures A1–A3 report half-year, detrended, and placebo-group event studies.

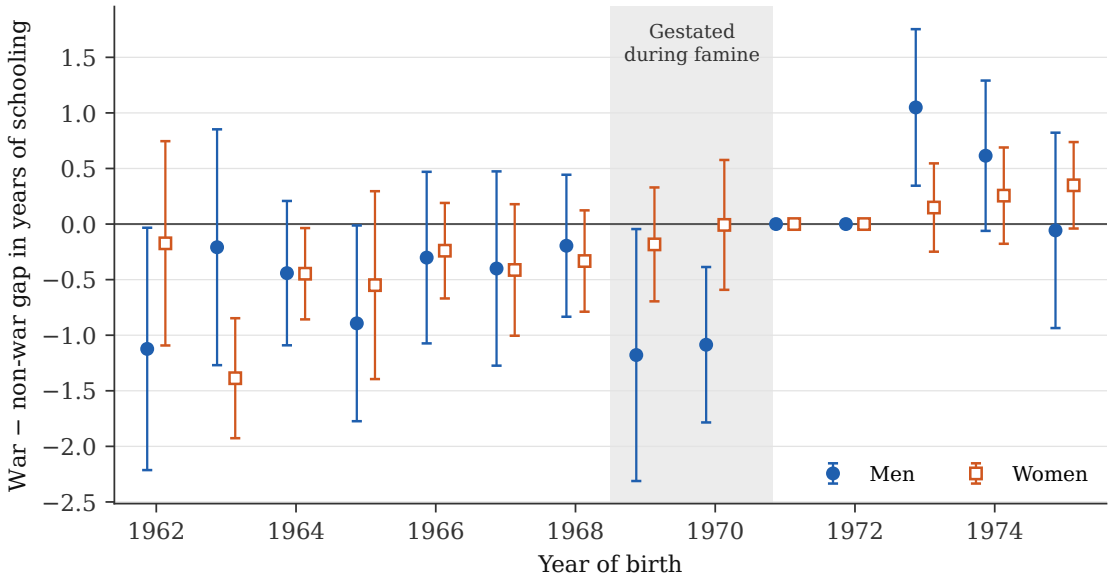


Figure 1. Event study: war–non-war gap in years of schooling, by year of birth.
Notes: Coefficients on war group × year of birth, relative to cohorts born 1971–72, estimated separately for men and women with the fixed effects in Eq. (1) and no trend. Bars are 95 percent confidence intervals clustered by ethnicity × birth year. Shading marks births with in-utero famine exposure (July 1968–October 1970). Exact birth dates, NDHS 2003–2024.

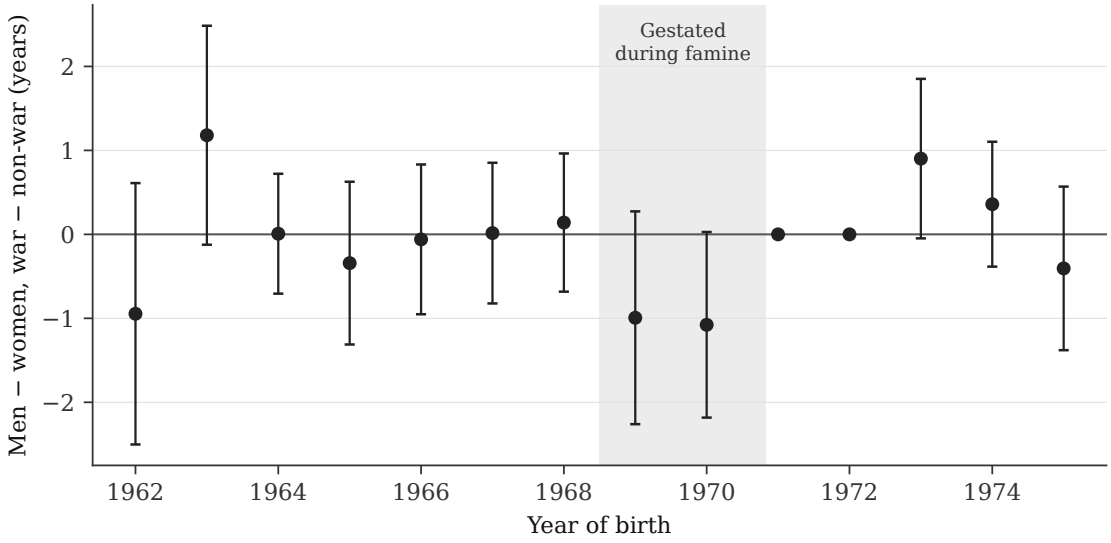


Figure 2. Triple difference: men minus women, by year of birth.
Notes: Coefficients on war group × male × year of birth, relative to 1971–72, from a stacked sample of men and women in which all fixed effects are interacted with sex and war group × year of birth fixed effects absorb the path common to both sexes. 95 percent confidence intervals clustered by ethnicity × birth year.

loss is concentrated at two margins: famine-exposed men are 6.6 percentage points less likely to have ever attended school and 12 percentage points less likely to reach secondary school, whereas the effect on completing primary school is small. Gestation during the pre-famine war months has no measurable effect on years of schooling. The famine–war contrast is significant for school entry (-12.7 points, $p = 0.02$) but imprecisely estimated for years of schooling ($p = 0.17$), reflecting the small size of the early-war cohort. Women gestated during the famine show no loss on any margin.[6]

Table 1. Famine gestation and schooling.

	Men				Women			
	Years	Ever attended	Primary or higher	Secondary or higher	Years	Ever attended	Primary or higher	Secondary or higher
Gestation in famine	−1.298***	−0.066**	−0.028	−0.120**	0.027	−0.006	−0.019	0.021
	(0.440)	(0.026)	(0.026)	(0.047)	(0.250)	(0.021)	(0.020)	(0.022)
Gestation in pre-famine war	0.039	0.061	0.217***	−0.116	1.054	0.051	0.067	0.092
	(1.033)	(0.062)	(0.078)	(0.103)	(0.855)	(0.071)	(0.075)	(0.077)
Famine − pre-famine war	−1.337	−0.127**	−0.245***	−0.004	−1.027	−0.057	−0.086	−0.071
	(0.978)	(0.056)	(0.069)	(0.097)	(0.775)	(0.058)	(0.063)	(0.071)
Mean	8.41	0.79	0.75	0.55	5.70	0.61	0.54	0.36
Observations	8,319	8,326	8,326	8,326	15,173	15,184	15,184	15,184

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Estimates of Eq. (1); coefficients are the effect of full (nine-month) gestation in the famine (June 1968–January 1970) and in the pre-famine war (July 1967–May 1968) for war-affected groups relative to other Nigerians born in the same month. All models include infancy exposure in each period, a war-group linear cohort trend, and fixed effects for birth year-month, ethnicity, state × round, war group × five-year age group, and war group × calendar month of birth. Standard errors clustered by ethnicity × birth year in parentheses. Means are weighted. Respondents born 1965–1973 with exact birth dates, NDHS 2003–2024; women born in this window are not interviewed in 2024.

Alternative specifications. Table 2 shows that the result does not hinge on specification choices. It is stable when imputed birth dates are included, the treated group is restricted to the Igbo, the war-group trend is omitted, standard errors are clustered by birth month, and the famine window is redefined (twelve windows yield -1.14 to -1.37). It remains negative when any single survey round is dropped.[7] Panel B confronts the UPE reform. Northern men born in 1969–70 have markedly more schooling than adjacent cohorts, which could masquerade as a relative loss for the war groups. Yet controlling for state UPE intensity (Osili and Long, 2008) interacted with cohort exposure leaves the estimate unchanged, and

[6]The other pre-specified outcomes are also unaffected (Online Appendix Table A2).

[7]The 2018 estimate is not statistically different from the others (Online Appendix Table A4).

absorbing every Northern cohort shock with Northern-ethnicity × birth-year fixed effects still yields a loss of 1.06 years ($p = 0.004$). The estimate is smaller and imprecise when only southern non-war groups serve as the comparison; these groups are, however, an imperfect counterfactual, since the Mid-West was itself a theater of war and the West had offered free primary schooling since 1955.[8] Panel C reports the triple difference, which nets out anything common to men and women of the same groups and cohorts. It yields a loss of 1.32 years ($p = 0.012$), nearly identical to the baseline.

Table 2. Alternative specifications: men's years of schooling.

	Gestation in famine	SE	Observations
Panel A. Specification and sample			
Main specification (Table 1)	−1.298***	(0.440)	8,319
Including imputed birth dates	−1.197***	(0.443)	9,798
Igbo only as treated group	−1.257***	(0.419)	7,769
No war-group trend	−1.773***	(0.478)	8,319
Clustering by birth year-month	−1.298**	(0.504)	8,319
Excluding the 2018 round	−0.836	(0.575)	
Twelve alternative famine windows	[−1.37, −1.14]		8,319
Panel B. Universal Primary Education and the comparison group			
+ UPE funds per capita × UPE exposure	−1.301***	(0.439)	8,319
+ High-UPE state × birth-year fixed effects	−1.380***	(0.450)	8,319
+ Northern ethnicity × birth-year fixed effects	−1.063***	(0.363)	8,319
+ 1976 state × birth-year fixed effects	−1.239**	(0.610)	8,319
Comparison group: southern non-war groups only	−0.236	(0.474)	4,738
Comparison group: Northern groups only	−2.260***	(0.703)	3,610
Panel C. Triple difference (male × war × famine gestation)			
Births 1965–1973	−1.324**	(0.516)	23,492
+ Northern ethnicity × birth-year fixed effects	−1.063**	(0.483)	23,492
+ 1976 state × birth-year fixed effects	−0.861	(0.549)	23,492
Births 1956–1986, with childhood war exposure	−1.466***	(0.477)	97,102

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Each row reports the effect of full gestation in the famine from a variant of Eq. (1). UPE intensity is the 1976 federal capital allocation for primary school construction per capita by state (Osili and Long, 2008, Table 1), mapped from current residence; UPE exposure is the number of primary-school-age years (6–11) on or after September 1976. Famine windows start in February, April, June, or August 1968 and end in September 1969, November 1969, or January 1970. In Panel C, men and women are stacked and all fixed effects are interacted with sex. Standard errors clustered by ethnicity × birth year.

Placebo inference. Differential cohort trends could, in principle, generate spurious dips. We therefore extend the sample to cohorts born 1956–1986 and re-estimate the model 74

[8] Online Appendix C discusses the comparison group in detail.

times, each time assigning a placebo 20-month famine to a period without war. Table 3 and Fig. 3 show that the actual estimate for men lies far outside the placebo distribution: none of the 74 placebo estimates is as large in absolute value. For women, the actual estimate sits squarely within its placebo distribution.

Table 3. Placebo inference.

	Men: years of schooling	Women: years of schooling
Actual famine estimate	−1.708	−0.240
Placebo estimates: mean	−0.055	0.074
(standard deviation)	(0.513)	(0.298)
Placebo estimates: 5th, 95th percentile	[−0.725, 0.760]	[−0.412, 0.534]
Number of placebo windows	74	74
Placebos at least as large in absolute value	0	30
Rank p-value	0.000	0.405

Notes: Cohorts born 1956–1986 with exact birth dates. Each placebo assigns a 20-month window whose gestation months do not overlap the war (July 1967–January 1970) and re-estimates Eq. (1), augmented with controls for real gestation, infancy, and childhood war exposure. The rank p-value is the share of placebo estimates at least as large in absolute value as the actual estimate.

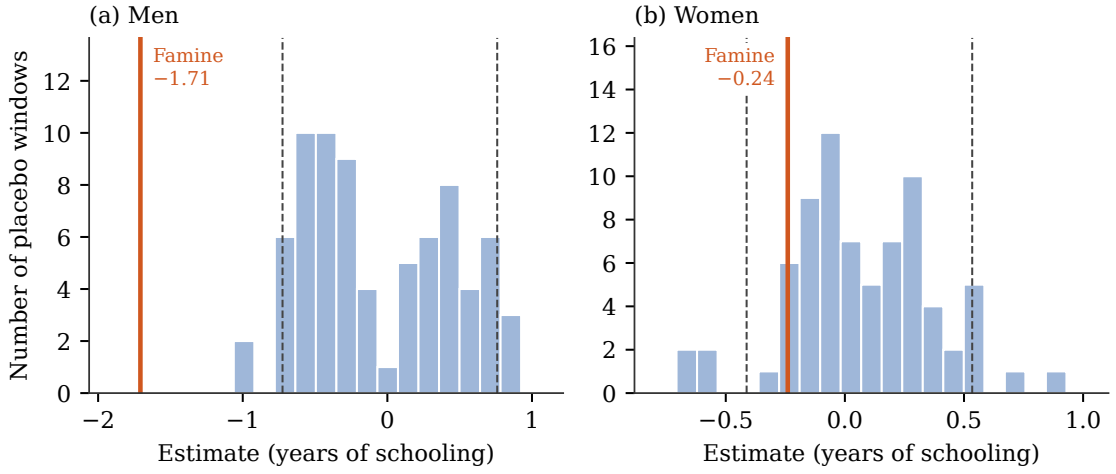


Figure 3. Distribution of placebo famine estimates.
Notes: Histograms of the 74 placebo estimates described in Table 3. Dashed lines mark the 5th and 95th percentiles; the solid line marks the actual famine estimate.

Selection. Fertility in the war-affected groups fell sharply during the famine, so mothers who gave birth may have differed from those who did not. The 1990 birth histories show no such selection: mothers of famine-gestated children in the former Eastern Region were similar in schooling, urban residence, age, and parity. Moreover, any selection of mothers

would apply equally to daughters, who show no loss.[9]

Margins and possible mechanisms. Table 4 sheds light on how the famine lowered attainment. Among men who completed primary school, famine exposure reduces the probability of continuing to secondary school by 11 percentage points. Famine-exposed men are less likely to read a full sentence, but not conditional on completed schooling, which suggests that the deficit operates through attainment rather than literacy at a given grade, although this decomposition is partly mechanical. Consistent with an earlier exit from school, they also marry about one year earlier. Taken together, the male-specific loss at school entry and at the secondary transition is consistent with evidence that maternal undernutrition particularly impairs male fetal development (Kraemer, 2000; Roseboom et al., 2006; Scholte et al., 2015). The children of these men live in poorer households, although a similar gap among children of fathers gestated in the early war indicates that this intergenerational pattern is not specific to the famine.[10]

Table 4. Margins of schooling and later outcomes (men).

	Famine gestation	Pre-famine war gestation	Mean
Completed primary | started school	0.040* (0.023)	0.188*** (0.058)	0.95
Reached secondary | completed primary	−0.110** (0.053)	−0.286*** (0.103)	0.74
Reads whole sentence	−0.092* (0.052)	−0.017 (0.110)	0.56
conditional on years of schooling	−0.003 (0.027)	0.014 (0.078)	0.56
Age at first marriage	−1.038** (0.452)	1.504 (1.269)	27.4
Ever married	0.044* (0.023)	0.150** (0.069)	0.96
Own household in top two wealth quintiles	−0.052 (0.047)	−0.029 (0.093)	0.48
Children: household in top two wealth quintiles	−0.121** (0.048)	−0.111 (0.122)	0.38

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Each row is a separate estimate of Eq. (1) for men born 1965–1973 with exact birth dates. "Conditional on years of schooling" adds fixed effects for each completed year. The literacy test is administered to respondents without secondary schooling; those with secondary schooling are coded as able to read. The last row is estimated at the child level for children aged 0–17 linked to co-resident fathers in the household roster (26,184 children), with additional fixed effects for round × child age × sex. Marriage outcomes were not pre-specified. Standard errors clustered by ethnicity × birth year in parentheses.

[9] See Online Appendix Tables A7–A8. Post-war school policies were permanent and applied to both sexes (Online Appendix C).

[10] See Online Appendix Table A10. The pre-famine war cohort is small (205 war-ethnicity men), which limits the precision of all famine–war comparisons.

6 Conclusion

This paper uses the timing of the Biafran blockade within the Nigerian Civil War to isolate the long-run effects of wartime famine. Men who were in utero during the famine completed 1.3 fewer years of schooling than neighboring cohorts, relative to other Nigerians born in the same months, while women were unaffected. The result withstands placebo famines, a triple difference against women, and controls for a contemporaneous national schooling reform. The costs of starvation in war thus extend for decades beyond the conflict itself. As sieges recur in contemporary wars, protecting civilian access to food yields returns long after the fighting ends.

Data availability

The NDHS data are available from The DHS Program (https://dhsprogram.com) upon registration. Replication code will be deposited in a public repository upon acceptance.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve readability, check grammar, and enhance clarity of expression in some sections of the manuscript. After using this tool, the authors thoroughly reviewed and edited all content and take full responsibility for the content of the published article.

Appendix A. Supplementary data

Supplementary material related to this article (Online Appendices A–C) can be found online.

References

Joseph B. Ajefu and Daniela Casale. The long-term effects of violent conflict on women's intra-household decision-making power. *Journal of Development Studies*, 57(10):1690–1709, 2021.

Mevlude Akbulut-Yuksel. Children of war: The long-run effects of large-scale physical destruction and warfare on children. *Journal of Human Resources*, 49(3):634–662, 2014.

Richard Akresh, Sonia Bhalotra, Marinella Leone, and Una Okonkwo Osili. War and stature: Growing up during the Nigerian civil war. *American Economic Review*, 102(3):273–277, 2012.

Richard Akresh, Sonia Bhalotra, Marinella Leone, and Una Okonkwo Osili. First- and second-generation impacts of the Biafran war. *Journal of Human Resources*, 58(2), 2023.

Douglas Almond and Janet Currie. Killing me softly: The fetal origins hypothesis. *Journal of Economic Perspectives*, 25(3):153–172, 2011.

Douglas Almond and Bhashkar Mazumder. Fetal origins and parental responses. *Annual Review of Economics*, 5:37–56, 2013.

Douglas Almond, Lena Edlund, Hongbin Li, and Junsen Zhang. Long-term effects of early-life development: Evidence from the 1959 to 1961 China famine. In Takatoshi Ito and Andrew Rose, editors, *The Economic Consequences of Demographic Change in East Asia*, pages 321–345. University of Chicago Press, 2010.

Mark Bray. *Universal Primary Education in Nigeria: A Study of Kano State*. Routledge and Kegan Paul, London, 1981.

Ralph Catalano and Tim Bruckner. Secondary sex ratios and male lifespan: Damaged or culled cohorts. *Proceedings of the National Academy of Sciences*, 103(5):1639–1643, 2006.

Marg Csapo. Universal primary education in Nigeria: Its problems and implications. *African Studies Review*, 26(1):91–106, 1983.

Janet Currie and Tom Vogl. Early-life health and adult circumstance in developing countries. *Annual Review of Economics*, 5:1–36, 2013.

Susanne R. de Rooij, Hans Wouters, Julie E. Yonker, Rebecca C. Painter, and Tessa J. Roseboom. Prenatal undernutrition and cognitive function in late adulthood. *Proceedings of the National Academy of Sciences*, 107(39):16881–16886, 2010.

Alex de Waal. *Mass Starvation: The History and Future of Famine*. Polity Press, Cambridge, 2017.

Stanley Diamond. The Ibo's plight. Letter, *The New York Review of Books*, February 24, 1972.

Lasse Heerten and A. Dirk Moses. The Nigeria–Biafra war: Postcolonial conflict and the question of genocide. *Journal of Genocide Research*, 16(2–3):169–203, 2014.

Martin Hult, Per Tornhammar, Peter Ueda, Charles Chima, Anna-Karin Edstedt Bonamy, Benjamin Ozumba, and Mikael Norman. Hypertension, diabetes and overweight: Looming legacies of the Biafran famine. *PLoS ONE*, 5(10):e13582, 2010.

Jeffrey R. Kling, Jeffrey B. Liebman, and Lawrence F. Katz. Experimental analysis of neighborhood effects. *Econometrica*, 75(1):83–119, 2007.

Sebastian Kraemer. The fragile male. *BMJ*, 321(7276):1609–1612, 2000.

Muhammad Farhan Majid. The persistent effects of in utero nutrition shocks over the life cycle: Evidence from Ramadan fasting. *Journal of Development Economics*, 117:48–57, 2015.

Xin Meng and Nancy Qian. The long term consequences of famine on survivors: Evidence from a unique natural experiment using China's Great Famine. NBER Working Paper 14917, National Bureau of Economic Research, 2009.

Sven Neelsen and Thomas Stratmann. Effects of prenatal and early life malnutrition: Evidence from the Greek famine. *Journal of Health Economics*, 30(3):479–488, 2011.

Okechukwu S. Ogah, Ayodipupo S. Oguntade, Innocent I. Chukwuonye, Ugochukwu U. Onyeonoro, Okechukwu O. Madukwe, Adanze Asinobi, Fisayo Ogah, Olanike A. Orimolade, Abiodun O. Babatunde, Maxwell F. Okeke, Ogochukwu P. Attah, Ifeoma G. Ebengho, Karen Sliwa, and Simon Stewart. Childhood and infant exposure to famine in the Biafran war is associated with hypertension in later life: The Abia NCDS study. *Journal of Human Hypertension*, 37(10):936–943, 2023.

Una Okonkwo Osili and Bridget Terry Long. Does female schooling reduce fertility? Evidence from Nigeria. *Journal of Development Economics*, 87(1):57–75, 2008.

Tessa Roseboom, Susanne de Rooij, and Rebecca Painter. The Dutch famine and its long-term consequences for adult health. *Early Human Development*, 82(8):485–491, 2006.

Jonathan Roth. Pretest with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–322, 2022.

Nicholas J. Sanders and Charles Stoecker. Where have all the young men gone? Using sex ratios to measure fetal death rates. *Journal of Health Economics*, 41:30–45, 2015.

Robert S. Scholte, Gerard J. van den Berg, and Maarten Lindeboom. Long-run effects of gestation during the Dutch Hunger Winter famine on labor market and hospitalization outcomes. *Journal of Health Economics*, 39:17–30, 2015.

Zena Stein, Mervyn Susser, Gerhart Saenger, and Francis Marolla. *Famine and Human Development: The Dutch Hunger Winter of 1944–1945*. Oxford University Press, New York, 1975.

Online Appendix

Starved in the womb: The Biafran blockade and men's schooling

Glory David Adebayo (Florida International University) and Jean Msabila (West Virginia University)

Online Appendix A: Tables and figures

Unless otherwise noted, estimates come from equation (1) of the main text: coefficients are the effect of full (nine-month) gestation in each period for war-affected groups (Igbo, Ijaw, Ibibio, Ekoi) relative to other Nigerians born in the same month, with infancy exposure, a war-group linear cohort trend, and fixed effects for birth year-month, ethnicity, state × survey round, war group × five-year age group, and war group × calendar month of birth. Samples use respondents born 1965–1973 with exact birth dates from the 2003–2024 NDHS. Standard errors are clustered by ethnicity × birth year. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A1: Descriptive statistics by period of birth

	Born before war (Jan 1965–Jul 1967)		War gestation (Aug 1967–Jun 1968)		Famine gestation (Jul 1968–Oct 1970)		Born after war (Nov 1970–1973)	
	War	Other	War	Other	War	Other	War	Other
Panel A. Men								
Years of schooling	9.77	8.26	10.16	7.09	10.12	9.12	10.23	7.64
Ever attended school	0.97	0.73	0.95	0.67	0.98	0.81	0.98	0.74
Secondary or higher	0.58	0.55	0.68	0.47	0.67	0.61	0.68	0.50
Age at survey	46.9	47.9	44.4	45.2	43.2	44.4	41.2	41.5
Observations	380	1,313	205	1,018	436	1,340	764	2,871
Panel B. Women								
Years of schooling	7.85	4.99	7.79	3.82	8.73	5.48	8.81	4.70
Ever attended school	0.85	0.52	0.87	0.43	0.89	0.57	0.91	0.51
Secondary or higher	0.48	0.32	0.47	0.23	0.57	0.35	0.57	0.29
Height (cm)	159.7	159.5	160.3	159.3	159.9	159.3	160.4	159.4
Observations	587	1,696	376	1,496	1,107	2,779	1,677	5,467

Notes: Weighted means. "War" denotes the Igbo, Ibibio, Ekoi, and Ijaw; "Other" denotes all other Nigerian groups. "War gestation": at least one gestation month in July 1967–May 1968 and none during the famine. "Famine gestation": at least one gestation month in June 1968–January 1970. Women born in this window are not interviewed in 2024. Among war-ethnicity men, schooling is flat across periods (10.16, 10.12, 10.23); the war–non-war gap narrows for famine-gestated cohorts because schooling among other men born in these months is higher, a swing concentrated among Northern cohorts born in 1969–70 (see Table A5 and Online Appendix C.3). Among war-ethnicity women, schooling rises from 7.79 to 8.73 while war-ethnicity men stay flat.

Table A2: Pre-specified primary and secondary outcomes

	Primary outcomes				Secondary outcomes		
	Men: schooling	Women: schooling	Women: height	Women: BMI	Women: overweight	Women: CEB	Men: CEB
Gestation in famine	-1.298^{***}	0.027	0.845^{**}	0.506	0.018	-0.085	0.269
	(0.440)	(0.250)	(0.383)	(0.356)	(0.039)	(0.195)	(0.324)
Gestation in pre-famine war	0.039	1.054	2.550^{**}	0.378	-0.005	0.061	-0.091
	(1.033)	(0.855)	(1.159)	(0.772)	(0.095)	(0.410)	(0.856)
Gestation in famine, alternative specifications							
Including imputed birth dates	-1.197^{***}	0.243	0.691^{*}	0.149	-0.002	-0.180	0.238
Igbo only as treated group	-1.257^{***}	0.017	0.343	0.502	0.013	-0.342^{**}	0.187
No war-group trend	-1.773^{***}	-0.115	0.304	0.193	-0.023	-0.042	0.134
Clustering by birth year-month	-1.298^{**}	0.027	0.845	0.506	0.018	-0.085	0.269
Placebo-in-time rank p	0.000	0.405	0.432				
Mean	8.37	5.59	159.4	24.5	0.38	5.78	5.54
Observations	8,319	15,173	12,553	12,543	12,543	15,184	8,326

Notes: CEB: children ever born. BMI: body mass index. Only men's schooling passes a Bonferroni threshold of 0.0125 for the four primary outcomes. The positive coefficient on women's height is larger for pre-famine war gestation than for famine gestation, falls without the war-group trend or among the Igbo, and lies inside its placebo distribution; we do not interpret it as a causal effect. Means are unweighted sample means.

Table A3: Robustness: sample, famine window, dose, and timing (men's years of schooling)

	Estimate	SE	p-value
Panel A. Dropping one survey round			
Main specification	−1.298	0.440	0.004
Drop 2003	−1.360	0.465	0.004
Drop 2008	−1.150	0.437	0.010
Drop 2013	−1.801	0.693	0.011
Drop 2018	−0.836	0.575	0.149
Drop 2024	−1.381	0.457	0.003
Panel B. Famine window: start month; end September 1969 / November 1969 / January 1970			
Start February 1968	−1.139 / −1.269 / −1.328		0.020 / 0.007 / 0.002
Start April 1968	−1.226 / −1.334 / −1.366		0.017 / 0.006 / 0.002
Start June 1968	−1.234 / −1.300 / −1.298		0.024 / 0.009 / 0.004
Start August 1968	−1.224 / −1.244 / −1.210		0.035 / 0.016 / 0.009
Panel C. Dose (months of gestation in famine, indicators)			
1–3 months	−1.336	0.470	0.005
4–6 months	−0.935	0.605	0.125
7–9 months	−1.011	0.403	0.014
Panel D. Trimester-specific exposure (full trimester)			
First trimester	−1.513	0.904	0.098
Second trimester	0.321	1.155	0.782
Third trimester	0.063	0.634	0.921

Notes: Panel B varies the famine window used to define gestation exposure; the June 1968–January 1970 cell is the main specification. Panels C and D replace the linear exposure measure. Dose and trimester estimates draw on the same timing variation as the sub-period estimates in Table A4 and are discussed in Online Appendix C.5.

Table A4: Estimates by survey round and by famine sub-period (men's years of schooling)

	Estimate	SE	Exposed war-ethnicity men
Panel A. Each survey round separately			
2003	−0.504	2.428	19
2008	−1.534	1.025	107
2013	−0.809	0.581	149
2018	−2.147***	0.730	94
2024	−0.413	1.126	67
Pooled: famine × 2018 interaction	−0.345	0.793	
Panel B. Famine sub-periods (full-gestation equivalents)			
First peak (June–December 1968)	−0.009	1.045	
Relief interlude (January–May 1969)	−1.783	1.360	
Second peak (June 1969–January 1970)	−1.318**	0.619	
Pre-famine war (July 1967–May 1968)	0.733	1.062	

Notes: Panel A estimates the model separately by round, without the war group × age-group term, which is collinear with cohort within a round. Share of war-ethnicity men born 1965–73 with imputed birth months: 18.4 percent (2003), 11.3 (2008), 5.4 (2013), 3.3 (2018), 0 (2024). Panel B replaces famine gestation with exposure in three sub-periods.

Table A5: Universal Primary Education and the comparison group

	(1) Main	(2) High-UPE state × exposure	(3) UPE funds p.c. (1953) × exposure	(4) UPE funds p.c. (1963) × exposure	(5) High-UPE state × birth-year FE	(6) North eth. × birth year FE	(7) 1976 state × birth year FE	(8) (3)+(6)
Panel A. Men's years of schooling ($N = 8{,}319$)								
Gestation in famine	-1.298^{***}	-1.301^{***}	-1.301^{***}	-1.298^{***}	-1.380^{***}	-1.063^{***}	-1.239^{**}	-1.067^{***}
	(0.440)	(0.440)	(0.439)	(0.440)	(0.450)	(0.363)	(0.610)	(0.360)
Famine − pre-famine	−1.337	−1.334	−1.338	−1.337	−1.131	−0.854	-1.977^{*}	−0.854
	(0.978)	(0.978)	(0.976)	(0.979)	(0.980)	(1.001)	(1.143)	(0.996)
UPE term		0.013	−0.010	−0.000				−0.015
		(0.070)	(0.034)	(0.056)				(0.035)
Panel B. Women's years of schooling ($N = 15{,}173$)								
Gestation in famine	0.027	0.016	0.033	0.035	−0.020	0.003	−0.381	0.009
	(0.250)	(0.249)	(0.253)	(0.254)	(0.245)	(0.274)	(0.303)	(0.283)
Panel C. Triple difference, male × famine gestation ($N = 23{,}492$)								
Estimate	-1.324^{**}		-1.333^{**}			-1.063^{**}	−0.861	
	(0.516)		(0.516)			(0.483)	(0.549)	
Panel D. Alternative comparison groups, men's years of schooling								
	Yoruba only		Southern non-war only		Northern groups only		Triple diff., southern only	
Gestation in famine	−0.325 (0.513)		−0.236 (0.474)		-2.260^{***} (0.703)		0.314 (0.484)	
Famine − pre-famine	-1.77^{*} (1.02)		−0.94 (0.98)		-2.66^{*} (1.40)			
Observations	2,942		4,738		3,610		13,618	

Notes: UPE intensity from Osili and Long (2008, Table 1): 1976 federal capital funds for primary school construction per capita by 1976 state, on a 1953-census or 1963-census population base (standardized), or an indicator for high-intensity (non-Western) states. Current states are mapped to 1976 states (Online Appendix B). UPE exposure is the number of primary-school-age years (ages 6–11) on or after September 1976. Columns 5–8 add the indicated fixed effects. In Panel C all fixed effects are interacted with sex. Southern non-war groups are the Yoruba, Mid-Western groups, and other non-Northern groups outside the war region.

Table A6: School-entry timing and within-war-ethnicity comparisons

	Gestation in famine	SE	Observations
Panel A. School-entry cohort controls (men's years of schooling)			
Main specification	−1.298***	(0.440)	8,319
+ War group × school-entry-year fixed effects	−1.080	(1.057)	8,319
+ War group × birth-year fixed effects	−0.320	(0.892)	8,319
Panel B. War groups only: famine enclave vs. early-captured areas			
Men, current residence	0.311	(0.866)	1,438
Men, current residence, excluding Anambra	0.809	(1.163)	1,226
Men, current residence, all birth dates	0.291	(0.821)	1,539
Men, state of birth (2024 only)	−1.229	(2.859)	234
Women, current residence	−0.219	(0.789)	3,188
Women's height, current residence	−0.646	(1.078)	2,541

Notes: Panel A: school-entry year is the calendar year of the first September at or after age six. Because the famine spans only 20 months, war group × birth-year fixed effects absorb most of the identifying variation. Panel B: war-affected groups only; "enclave" is residence in Imo, Abia, or Anambra (under blockade June 1968–January 1970); "early-captured" is residence in Enugu, Ebonyi, Cross River, Akwa Ibom, Rivers, or Bayelsa. The enclave indicator replaces the war indicator in equation (1). Standard errors clustered by state × birth year (81 clusters). These comparisons are imprecise: confidence intervals include both zero and the main estimate.

Table A7: Characteristics of mothers of famine-gestated births (1990 birth histories)

	Famine gestation	Pre-famine war gestation	Mean
Mother's years of schooling	−0.214 (0.221)	0.040 (0.365)	1.58
Mother ever attended school	−0.074* (0.042)	−0.030 (0.069)	0.27
Mother has secondary schooling	0.001 (0.016)	0.019 (0.026)	0.06
Mother lives in urban area	−0.000 (0.026)	0.061 (0.049)	0.31
Mother's age at birth	0.006 (0.346)	−0.715 (0.568)	21.4
Birth order	−0.002 (0.159)	−0.191 (0.252)	2.63
First birth	0.039 (0.045)	0.104** (0.048)	0.33
Child is male	0.002 (0.052)	0.069 (0.094)	0.55

Notes: NDHS 1990 birth histories, births 1965–1973 (N = 4,218; 276 births to mothers in the former Eastern Region with famine gestation). The 1990 round is the only one in which mothers of these births are still of reproductive age; it does not record ethnicity, so the treated group is residence in the former Eastern Region (Anambra, Imo, Cross River, Akwa Ibom, Rivers). Each row regresses the characteristic on full famine and pre-famine war gestation exposure interacted with the treated group, a treated-group trend, and fixed effects for birth year-month, state, and treated group × calendar month. Standard errors clustered by state × birth year. Mothers were aged 29 or younger at a 1969 birth. Taking the −7.4 point difference in ever attending school at face value, a generous intergenerational coefficient implies that maternal composition could account for at most 0.1–0.2 of the 1.3-year effect.

Table A8: Fertility, sex ratio at birth, infant mortality, and adult survival

	Pre-war (1965–Jun 67)	War, pre-famine (Jul 67–May 68)	Famine (Jun 68–Jan 70)	Conceived in famine (Feb–Oct 70)	Post-war (Nov 70–73)
Panel A. Birth histories, by quarter of birth (period averages)					
Birth rate ratio, war / other groups	1.23	1.32	0.96	0.88	1.12
Male share of births, war groups	0.559	0.590	0.516	0.506	0.558
Male share of births, other groups	0.598	0.521	0.513	0.600	0.551
Infant mortality per 1,000, war groups	113	96	149	139	137
Infant mortality per 1,000, other groups	159	157	187	140	130
War-group births in sample	214	106	141	137	590

Panel B. Adult sex ratio (male share), war − other groups, relative to other years	Births 1960–1980	Births 1964–1976
Birth years 1967–68	-0.045^{***} (0.013)	-0.039^{**} (0.015)
Birth years 1969–70	−0.022 (0.015)	−0.017 (0.015)
Observations	110,426	64,333

Notes: Panel A: NDHS birth histories; war group defined by residence in the former Eastern Region in 1990 and by mother's ethnicity in 2003 and 2008. Birth rates are births per 1,000 women aged 15–44 per quarter; quarterly war-group counts are small (15–60 births). The male share is volatile in both groups, and the high pre-famine male share among war-group births is consistent with the omission of daughters from long recall histories; we therefore do not interpret the sex ratio as evidence of selective male fetal loss. Panel B: de jure household members; regression of male on war group × birth-year-group indicators with birth-year, ethnicity, state × round, and war group × round fixed effects and a war-group trend; standard errors clustered by ethnicity × birth year. Birth years 1967–68 include some famine-gestated births from the second half of 1968.

Table A9: Mechanisms: schooling margins, literacy, health, and adult outcomes (men)

	Famine	Pre-famine war	Mean	N
Panel A. Schooling margins				
Ever attended school	−0.066** (0.026)	0.061 (0.062)	0.79	8,326
Completed primary | started school	0.040* (0.023)	0.188*** (0.058)	0.95	6,569
Reached secondary | completed primary	−0.110** (0.053)	−0.286*** (0.103)	0.74	6,207
Years of schooling | started school	−0.679* (0.396)	−0.541 (0.931)	10.6	6,562
Panel B. Literacy and information				
Reads whole sentence	−0.092* (0.052)	−0.017 (0.110)	0.56	8,261
conditional on years of schooling	−0.003 (0.027)	0.014 (0.078)	0.56	8,254
Reads any part of sentence	−0.090** (0.040)	−0.126 (0.094)	0.73	8,261
conditional on years of schooling	−0.025 (0.037)	−0.174** (0.074)	0.73	8,254
Any media weekly	−0.077 (0.049)	−0.108 (0.099)	0.70	8,326
Reads newspaper weekly	−0.080* (0.044)	−0.028 (0.112)	0.26	8,326
Panel C. Health (2024 only)				
Ever told hypertensive	0.002 (0.105)	−0.089 (0.298)	0.17	1,209
Ever told diabetic	−0.094 (0.091)	−0.441* (0.233)	0.04	1,209
Any disability	−0.038 (0.049)	−0.086 (0.168)	0.04	1,209
Any functional difficulty	0.078 (0.140)	−0.404 (0.292)	0.30	1,209
Panel D. Adult economic position				
Household in top two wealth quintiles	−0.052 (0.047)	−0.029 (0.093)	0.48	8,326
Currently working	0.016 (0.021)	0.052 (0.047)	0.97	8,317
Owns land	0.063 (0.060)	−0.249* (0.146)	0.62	5,805
Has bank account	0.033 (0.076)	0.345* (0.176)	0.55	2,989
Lives in urban area	0.027 (0.037)	−0.006 (0.101)	0.42	8,326
Lifelong resident of current place	0.013 (0.057)	−0.372*** (0.112)	0.55	5,481
Panel E. Family formation (not pre-specified)				
Age at first marriage	−1.038** (0.452)	1.504 (1.269)	27.4	8,014
Ever married	0.044* (0.023)	0.150** (0.069)	0.96	8,326

Notes: "Conditional on years of schooling" adds fixed effects for each completed year of schooling; it is a decomposition of the total effect, not a control. The DHS literacy test is administered to respondents without secondary schooling; those with secondary schooling are coded as able to read, so the conditional estimate is partly mechanical. The pre-famine column shows large offsetting movements at the primary and secondary margins for the small early-war cohort (205 war-ethnicity men), with no net effect on years of schooling, and a lower probability of lifelong residence, consistent with displacement during the early war.

Table A10: Second generation: children of war-ethnicity fathers

	Household in top two wealth quintiles	Child index (SD)	Attending school (all)	Attending school (daughters)
Father: gestation in famine	−0.121**	−0.127*	−0.085*	−0.101**
	(0.048)	(0.067)	(0.047)	(0.049)
Father: gestation in pre-famine war	−0.111	0.023	0.038	0.080
	(0.122)	(0.151)	(0.092)	(0.105)
Father: gestation in famine, alternative specifications				
Igbo fathers only	−0.122***	−0.152**	−0.109**	−0.115**
	(0.040)	(0.071)	(0.049)	(0.050)
Controlling for father's schooling	−0.064		−0.035	
	(0.049)		(0.042)	
Fathers with all birth dates	−0.138***	−0.119*	−0.073	
	(0.050)	(0.063)	(0.045)	
Placebo-in-time rank p (74 windows)	0.027	0.108	0.149	
Mean	0.38	0.08	0.71	0.70
Observations (children)	26,184	23,134	16,001	7,627

Notes: Children aged 0–17 in NDHS 2003–2024 household rosters linked to co-resident fathers born 1965–1973 with exact birth dates (26,186 children, 987 of them children of 324 famine-exposed war-ethnicity fathers; two lack household wealth). Linked fathers reproduce the main result for their own schooling (−1.31, SE 0.56). Regressions use the father's exposure variables and fixed effects plus round × child age × child sex fixed effects. The child index averages standardized school attendance (ages 6–17), ever attending school (7–17), not being two or more grades behind (8–17), and height-for-age (under 5), each standardized by age using children of non-war fathers (Kling et al., 2007). Sons: attendance −0.053 (0.050). Because the father's pre-famine war coefficient on wealth is nearly as large as the famine coefficient, the wealth gap is not specific to the famine. When exposure is measured through co-resident mothers (37,987 children), children's outcomes respond as much to the mother's pre-famine war exposure as to her famine exposure, and we do not interpret them. The child-level wealth estimate is larger than the men's own estimate in Table A9 because the child sample is restricted to fathers living with their children and weights households by number of children.

Table A11: Re-estimating the ethnicity-by-cohort design of prior work

	(1) Akresh-style (control to 1974)	(2) + War × age-group FE (pooled)	(3) + War-group trend	(4) Placebo: fake war 1971–75	(5) Placebo: North as treated
Women: height (cm)	−0.647***	−0.263	0.328	−0.283	−0.090
	(0.226)	(0.300)	(0.379)	(0.229)	(0.267)
Women: years of schooling	−0.468***	−0.494***	0.236	−0.617***	0.014
	(0.155)	(0.171)	(0.182)	(0.141)	(0.186)
Women: widowed	0.015	−0.005	−0.028***	0.020***	−0.009
	(0.010)	(0.009)	(0.010)	(0.006)	(0.007)
Women: age at first marriage	−0.691***	−0.533*	−0.123	−0.429**	−0.090
	(0.239)	(0.274)	(0.358)	(0.214)	(0.165)
Women: physical/sexual violence	−0.008	0.014	−0.003	0.010	−0.020
	(0.015)	(0.019)	(0.023)	(0.017)	(0.012)
Men: years of schooling	−0.499**	−0.823***	−0.814***	−0.132	−0.368*
	(0.228)	(0.229)	(0.280)	(0.195)	(0.221)
Men: age at first marriage	0.568*	0.331	0.029	0.781***	−0.532**
	(0.333)	(0.354)	(0.385)	(0.261)	(0.257)
Men: polygynous	−0.009	0.022	0.040**	−0.053***	0.025
	(0.014)	(0.015)	(0.018)	(0.015)	(0.021)

Notes: Coefficient on war group × exposed cohort (born before November 1970). Column 1: fixed effects for birth year, ethnicity, state, and round; control cohorts born November 1970–1974, close to Akresh et al. (2012). Columns 2–3: control cohorts born November 1970–1980, adding war group × five-year age-group fixed effects (identified by pooling rounds) and then a war-group linear cohort trend. Column 4: post-war cohorts only, "exposed" cohorts born 1971–75 versus 1976–85; no cohort in this sample was exposed. Column 5: war groups dropped; Hausa, Fulani, and Kanuri as fake-treated. Child neonatal mortality is 1.4 (S0) and 1.3 (with war × age-group FE) percentage points higher for children of mothers exposed in utero, both $p < 0.01$.

Table A12: Event-study averages by period of birth (years of schooling)

	Born before war	War gestation	Famine gestation	Born after war
Panel A. Cohorts born 1962–1975, reference 1971–72, no trend				
War groups, men	−0.43 (0.35)	−0.28 (0.31)	−0.95 (0.34)	0.53 (0.29)
War groups, women	−0.61 (0.24)	−0.38 (0.25)	−0.19 (0.22)	0.21 (0.18)
North as fake-treated, men	−0.81 (0.48)	0.02 (0.44)	0.40 (0.37)	−0.10 (0.39)
Triple difference, men − women	0.15 (0.40)	0.07 (0.40)	−0.79 (0.40)	0.29 (0.34)
Panel B. Cohorts born 1958–1986, trend fitted on never-exposed anchors				
	Born 1958–66	War gestation	Famine gestation	Never exposed, out of sample
War groups, men	−0.13 (0.54)	−0.22 (0.40)	−0.87 (0.36)	0.16 (0.14)
War groups, women	−0.14 (0.31)	−0.14 (0.27)	−0.02 (0.23)	0.16 (0.07)
North as fake-treated, men	−1.40 (0.63)	−0.34 (0.47)	0.22 (0.44)	0.54 (0.22)

Notes: Averages of half-year event-study coefficients within each period, with standard errors from the clustered covariance matrix. Panel A: Figure A1. Born after war: 1973–75. In the triple difference, the famine-gestated average minus the average of all earlier cohorts is −0.92 (SE 0.42). Panel B: Figure A2; the war-group trend is pinned by never-exposed anchor cohorts born 1971–72 and 1981–82, and the last column averages the remaining never-exposed cohorts (1973–80, 1983–86). War men's and war women's out-of-sample deviations are identical (+0.16), so the misfit of the linear trend is common to both sexes and is differenced out in the triple difference. Joint Wald tests that all out-of-sample coefficients are zero reject in every panel, including the placebo group; with 24 clustered restrictions these tests over-reject, so we report averages.

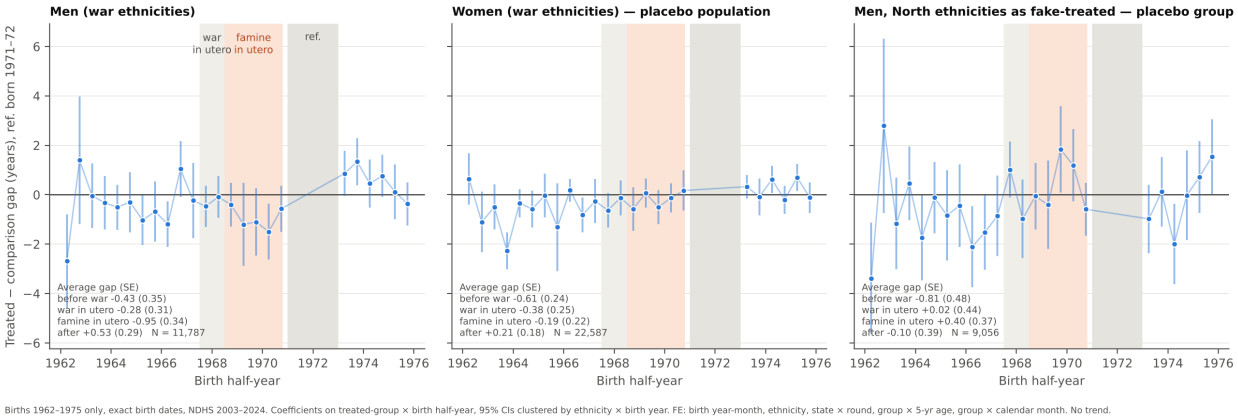


Figure A1: Half-year event studies, cohorts born 1962–1975

Notes: Coefficients on treated group × birth half-year, relative to 1971–72, with 95 percent confidence intervals. Left: war groups, men. Middle: war groups, women (placebo population). Right: Northern groups as fake-treated, men, war groups excluded (placebo group). Shading: in-utero war exposure before the famine (grey) and in-utero famine exposure (orange).

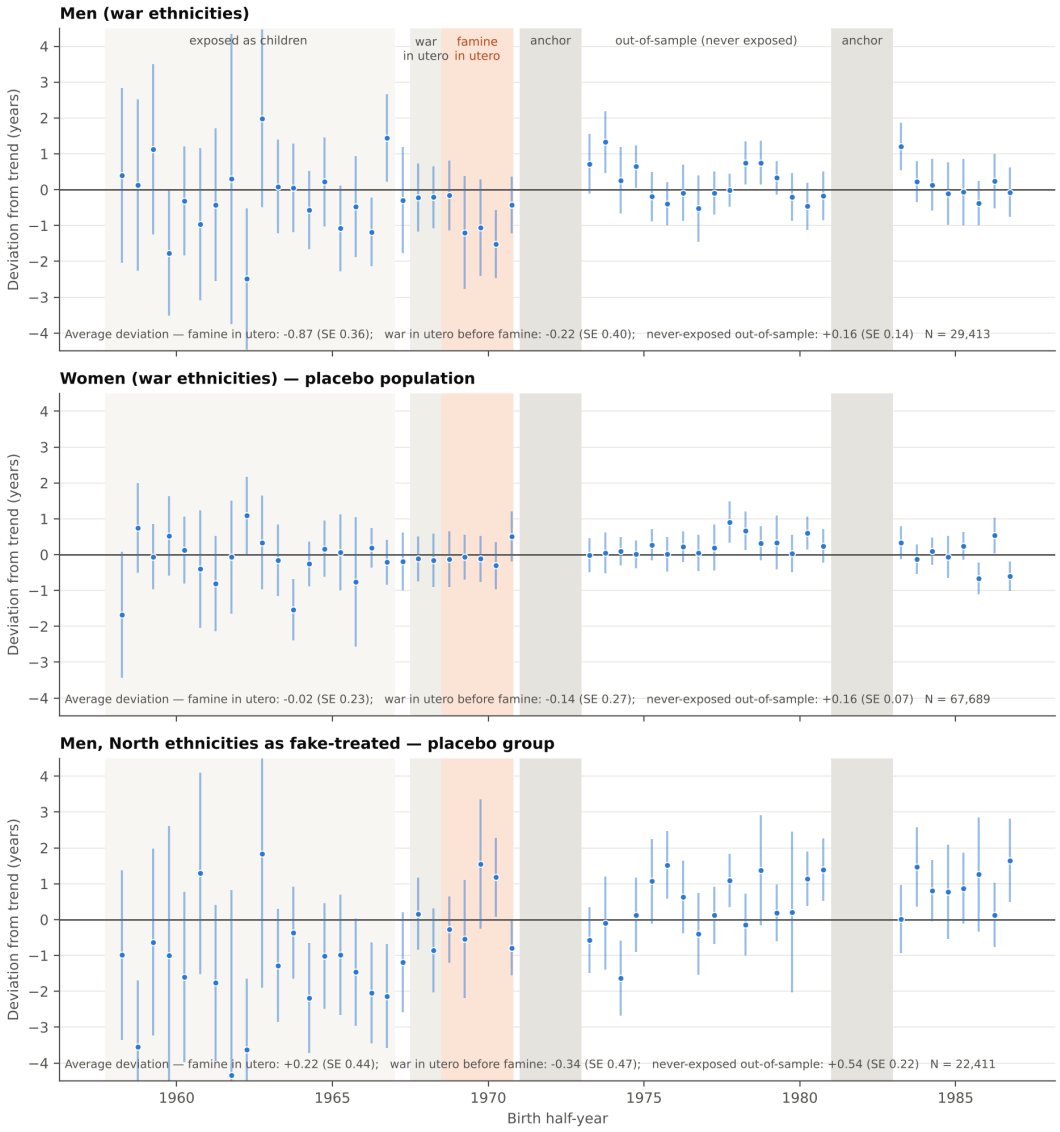


Figure A2: Detrended event studies, cohorts born 1958–1986

Notes: Coefficients on treated group × birth half-year with a treated-group linear trend pinned by never-exposed anchor cohorts (1971–72 and 1981–82). Coefficients between and after the anchors are out-of-sample checks of the trend. See Table A12, Panel B.

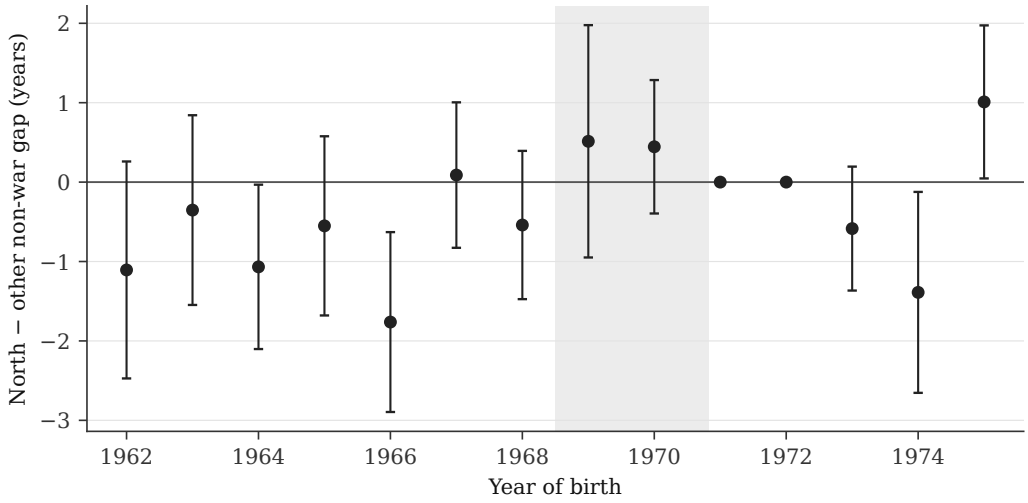


Figure A3: Placebo group: Northern ethnic groups as fake-treated, by year of birth
Notes: Same specification as Figure 1 of the main text, men, war groups excluded, Hausa, Fulani, and Kanuri treated as if exposed. There is no famine-period dip.

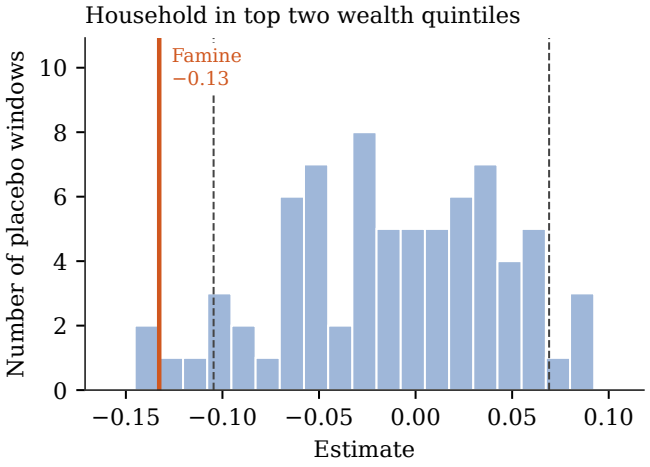


Figure A4: Placebo famines: second-generation household wealth
Notes: 74 placebo estimates for children's household wealth (top two quintiles) by the father's placebo famine exposure; the solid line marks the actual estimate (−0.133; 2 of 74 placebos at least as large). Dashed lines: 5th and 95th percentiles.

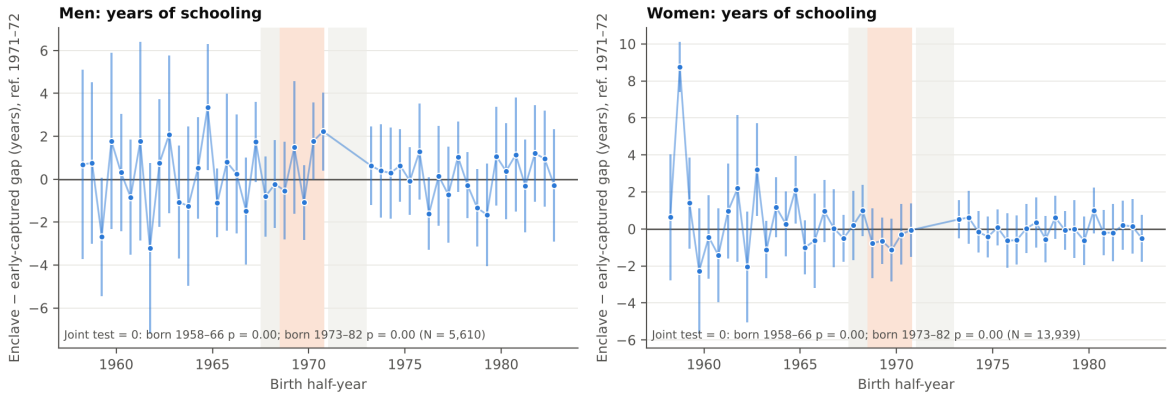


Figure A5: War groups only: famine enclave versus early-captured areas

Notes: Coefficients on residence in the famine enclave (Imo, Abia, Anambra) × birth half-year, relative to residence in areas captured before June 1968, war groups only, relative to 1971–72. See Table A6, Panel B.

Online Appendix B: Data, variables, and definitions

Surveys. Individual (women's and men's) recode files, household member recodes, and birth recodes of the Nigeria DHS 2003, 2008, 2013, 2018, and 2024; birth recode of the 1990 NDHS. The 1999 round is not publicly available. All estimates use DHS sampling weights, normalized to mean one within each round (and within sex in stacked samples).

Ethnicity. Detailed ethnicity is recorded in country-specific variables in 2003 (women `s118`, men `sm130`) and 2008 (`s119`/`sm119`), in the standard variable `v131` in 2013 and 2024, and only as ten broad groups in 2018. Detailed codes are mapped to broad groups using a common crosswalk. The treated ("war") group comprises the broad groups Igbo, Ijaw, Ibibio, and Ekoi; all other groups form the comparison group. Northern groups (Hausa, Fulani, Kanuri, and other Northern groups) are used as a placebo treated group in Table A11 and Figure A3. "Southern non-war groups" are the Yoruba, Mid-Western groups, and other non-Northern groups outside the war region.

Date of birth and exposure. Birth month is in century-month code (CMC). Imputed dates are identified from the DHS completeness-of-information variable; the main sample excludes them. Gestation is the nine months $b-9,\ldots,b-1$; infancy is months $b,\ldots,b+23$. Famine: June 1968 (CMC 822) to January 1970 (CMC 841). Pre-famine war: July 1967 (CMC 811) to May 1968 (CMC 821). Exposure variables count overlapping months (0–9 for gestation, 0–20 for infancy). The extended sample for placebo tests (births 1956–1986) adds childhood war exposure, the number of months between ages 2 and 16 spent in the war.

Outcomes. *Years of schooling*: completed years (`v133`/`mv133`). *Ever attended school*: highest level above none. *Primary or higher* and *secondary or higher*: highest level completed or attended (DHS education level variable). *Reads whole sentence*: DHS literacy variable equal to 2; respondents with secondary schooling are coded as able to read. *Age at first marriage*: DHS age at first cohabitation. *Household in top two wealth quintiles*: DHS wealth index quintile 4 or 5. Women's height and BMI: DHS anthropometry. Health outcomes in 2024: self-reported diagnosis of hypertension or diabetes; disability and functional difficulty from the household roster (Washington Group questions).

UPE intensity. Osili and Long (2008, Table 1) report 1976 federal capital funds for primary school construction per capita for the nineteen 1976 states. Current states are mapped to 1976 states as follows: Oyo (Oyo, Osun), Ogun, Ondo (Ondo, Ekiti), Lagos, Anambra (Anambra, Enugu, Ebonyi), Imo (Imo, Abia), Rivers (Rivers, Bayelsa), Cross

River (Cross River, Akwa Ibom), Bendel (Edo, Delta), Kwara (Kwara, Kogi), Benue, Plateau (Plateau, Nasarawa), Niger (Niger, FCT), Kaduna (Kaduna, Katsina), Kano (Kano, Jigawa), Sokoto (Sokoto, Kebbi, Zamfara), Bauchi (Bauchi, Gombe), Borno (Borno, Yobe), Gongola (Adamawa, Taraba). High-intensity states are all except Oyo, Ogun, Ondo, and Lagos. UPE exposure is the number of primary-school-age years (6–11) on or after September 1976, computed from month of birth.

Second-generation linkage. Children aged 0–17 in the household roster are linked to their co-resident father through the father's line number (`hv114`) and the men's interview (cluster, household, line), which provides the father's exact date of birth and ethnicity. Child outcomes: currently attending school (`hv121`, ages 6–17), ever attended (ages 7–17), two or more grades behind expected grade (ages 8–17), height-for-age z-score (`hc70`, under 5), and household wealth quintile (`hv270`). Mother linkage uses `hv112` and the women's interview.

Mother-selection sample. 1990 birth histories, births 1965–1973; the treated group is residence in the former Eastern Region (Anambra, Imo, Cross River, Akwa Ibom, Rivers), because the 1990 survey does not record ethnicity.

Placebo famines. Each placebo is a 20-month window starting at CMC s, for $s \in \{692, 695, \ldots, 779\} \cup \{860, 863, \ldots, 989\}$, so that its gestation months do not overlap the war; windows with fewer than 50 exposed war-group observations are dropped, leaving 74. Each placebo regression includes the real gestation, infancy, and childhood exposure terms.

Online Appendix C: Background and extended discussion

C.1 Chronology

Table A13: Chronology of the war and the famine

Date	Event
30 May 1967	Eastern Region declares independence as the Republic of Biafra
6 July 1967	Fighting begins (start of pre-famine war period)
July 1967	Federal forces take Nsukka
October 1967	Federal forces take Enugu (4 Oct) and Calabar
January–March 1968	Awka (Jan) and Onitsha (Mar) fall
April–June 1968	Ikot Ekpene, Abakaliki, and Afikpo fall
May 1968	Port Harcourt falls; Biafra loses its last major seaport
June 1968	Near-total blockade of the enclave (start of famine period)
16 August 1968	Relief agencies describe the greatest emergency since the Second World War
September–October 1968	Aba, Owerri, and Okigwe fall; Owerri is retaken by Biafra in April 1969
5 June 1969	Red Cross relief aircraft shot down; relief flights suspended for a period
Second half of 1969	Reports of child deaths in the hundreds to over a thousand per day
January 1970	Uli airstrip and remaining enclave fall
15 January 1970	Biafra surrenders (end of famine period)
1970–71	East Central State takes over all schools
September 1976	Universal Primary Education launched nationally

C.2 Post-war schooling policy

The East Central State Public Education Edict of 1970–71 brought all schools in the state under government control; the number of functioning secondary schools fell from about 290 before the war to about 190 in the early 1970s (Diamond, 1972). UPE abolished primary school fees in September 1976 (Csapo, 1983; Bray, 1981); Osili and Long (2008) classify the Eastern and Northern states as high-intensity and the former Western Region, which had free primary schooling from 1955, as low-intensity. Both policies were permanent and applied to boys and girls. A permanent policy cannot by itself produce a male-specific dip confined to famine-gestated cohorts that reverses for later cohorts. Famine-gestated boys would have entered school in 1975–76; adding war group $\times$ school-entry-year fixed effects leaves the point estimate largely intact but imprecise (Table A6), so the timing of school entry cannot be fully separated from famine gestation statistically.

C.3 The comparison group

Raw schooling among Northern men rises from 3.8–5.0 years for cohorts born 1962–68 to 6.5 and 6.2 years for cohorts born in 1969 and 1970, before falling back to 4.0–4.9 years for 1971–73 births; Northern women born in 1969 show a similar rise. These cohorts were among the first to enter school under UPE, whose enrollment gains were largest in the North. Table A5 keeps all non-war groups, as specified before estimation, and controls for UPE directly. The famine estimate is unchanged by state UPE intensity interacted with exposure and remains −1.06 when all Northern cohort shocks are absorbed. With only southern non-war groups as the comparison, the estimate is −0.24 (SE 0.47); with only Yoruba, −0.33 (SE 0.51); with only Northern groups, −2.26 (SE 0.70). Southern groups are not an unaffected comparison: the Mid-West was invaded by Biafran forces in August 1967 and was a theater of war, and the West's earlier free primary schooling gives it a different trajectory around 1976. The Northern-ethnicity × birth-year specification addresses the concern that motivates dropping the North while retaining Northern observations for the other fixed effects. The southern-only estimates are nonetheless much smaller, with confidence intervals that include both zero and about −1.2 years; readers who regard southern groups as the appropriate counterfactual should place less weight on the main estimate.

C.4 Selection

Fertility in the war groups fell during the famine (Table A8). Mothers of famine-gestated births did not differ observably from other mothers (Table A7), and any selection of mothers would also affect daughters, who show no loss. A fall in the share of male births is a common signature of fetal stress (Catalano and Bruckner, 2006; Sanders and Stoecker, 2015), but here the male share is volatile in both groups and the pre-famine level among war groups is implausibly high, so we do not claim that selective male fetal loss biases the estimates toward zero. The adult sex-ratio deficit among war groups is no larger for the cohorts most exposed to the famine (Table A8, Panel B).

C.5 Timing within the famine

The dose–response pattern in months of famine gestation is not monotone, and exposure confined to the first trimester comes mainly from births in August–October 1970 (conceived during the second famine peak), whereas exposure confined to the third trimester comes from births in July–September 1968 (the first peak). Consistently, gestation during the first peak has no effect, while gestation during the relief interlude and the second peak have

negative coefficients (Table A4). One reading emphasizes early-gestation sensitivity, as in the Dutch Hunger Winter and Ramadan studies (Roseboom et al., 2006; Majid, 2015); another emphasizes cumulative depletion of maternal reserves as the blockade persisted. The data cannot distinguish between them.

C.6 The famine–war contrast

In the main specification, the difference between famine and pre-famine war gestation is −1.34 years (SE 0.98, $p = 0.17$) for years of schooling and −12.7 points ($p = 0.02$) for ever attending school. In the triple difference, male pre-famine war gestation has a coefficient of −1.02 years ($p = 0.42$) and −20 points for reaching secondary school ($p = 0.08$). The pre-famine war cohort is small (205 war-ethnicity men), so the evidence shows a loss concentrated among famine-gestated cohorts without establishing that gestation during the earlier war months was harmless.

C.7 Mechanisms

The male-specific loss at school entry and at the transition to secondary school is consistent with the fetal-origins literature (Almond and Currie, 2011; Currie and Vogl, 2013) and with the greater vulnerability of male fetuses (Kraemer, 2000). The absence of a literacy gap at a given level of schooling resembles evidence from the Dutch Hunger Winter, where gestational exposure did not affect the measured intelligence of male conscripts (Stein et al., 1975) but affected later labor-market outcomes and, for early-gestation exposure, late-life cognition (Scholte et al., 2015; de Rooij et al., 2010). Candidate channels include health and stamina, or parental investment responding to children's early endowments (Almond and Mazumder, 2013); the data do not observe these directly.

C.8 Revisiting prior estimates

Pooling survey rounds permits two checks that single cross-sections do not (Table A11). A placebo war assigned to cohorts born 1971–75, none of whom were exposed, produces statistically significant "effects" for women's schooling, widowhood, age at first marriage of women and men, and men's polygyny, indicating differential cohort trajectories between war and non-war groups. Allowing war groups their own age profiles and cohort trend moves the women's height estimate from −0.65 cm to +0.33 cm and changes the sign of the women's schooling estimate. Men's schooling is an exception: its estimate is stable across specifications and its fake-war placebo is small. Two measurement issues also matter: 30–34

percent of exposed war-ethnicity women in 2003 and 2008 have imputed birth months, and the sibling histories used to build ethnic war-mortality intensity show strong age heaping and no detectable excess mortality in 1967–69.